CALCULUS

CK-12 Foundation

Default Text

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CHAPTER OVERVIEW

1: Limit

1.1: Concept of Limit

When learning about the **end behavior** of a rational function you described the function as either having a **horizontal asymptote** at zero or another number, or going to infinity. **Limit notation** is a way of describing this end behavior mathematically.

You already know that as x gets extremely large then the function $f(x)=\frac{8x^4+4x^3+3x^2-10}{3x^4+6x^2+9x}$ goes $^8/_3$ because the greatest power are equal and $^8/_3$ is the ration of the leading coefficients. How is this statement represented using limit notation?

Introduction to Limits

Limit notation is a way of stating an idea that is a little more subtle than simply saying x=5 or y=3.

$$\lim_{x\to a} f(x)=b$$

"The limit of f of x as x approaches a is b"

The letter a can be any number or infinity. The function f(x) is any function of x. The letter b can be any number. If the function goes to infinity, then instead of writing "=∞" you should write that the limit does not exist or "*DNE*". This is because infinity is not a number. If a function goes to infinity then it has no limit.

Take the following limit:

The limit of $y=4x^2$ as x approaches 2 is 16

In limit notation, this would be:

$$\lim_{x\to 2} 4x^2 = 16$$

While a function may never actually reach a height of b it will get arbitrarily close to b. One way to think about the concept of a limit is to use a physical example. Stand some distance from a wall and then take a big step to get halfway to the wall. Take another step to go halfway to the wall again. If you keep taking steps that take you halfway to the wall then two things will happen. First, you will get extremely close to the wall but never actually reach the wall regardless of how many steps you take. Second, an observer who wishes to describe your situation would notice that the wall acts as a limit to how far you can go.

Examples

Example 1

Earlier, you were asked how to write the statement "The limit of $f(x)=\frac{8x^4+4x^3+3x^2-10}{3x^4+6x^2+9x}$ as x approaches infinity is $^8/_3$" in limit notation.

This can be written using limit notation as:

$$\lim_{x\to\infty}\left(\frac{8x^4+4x^3+3x^2-10}{3x^4+6x^2+9x}\right)=\frac{8}{3}$$

Example 2

Translate the following mathematical statement into words.

$$\lim_{n\to\infty}\sum_{i=1}^{n}\left(\frac{1}{2}\right)^i=1$$

The limit of the sum of $^1/_2+^1/_4+^1/_8+\cdots$ as the number of terms approaches infinity is 1.

Example 3

Use limit notation to represent the following mathematical statement.

$^1/_3+^1/_9+^1/_{27}+\cdots=^1/_2$

$$\lim_{n\to\infty}\sum_{i=1}^{n}\left(\frac{1}{3}\right)^i=\frac{1}{2}$$

Example 4

Describe the end behavior of the following rational function at infinity and negative infinity using limits.

$$f(x)=\frac{-5x^3+4x^2-10}{10x^3+3x^2+98}$$

Since the function has equal powers of x in the numerator and in the denominator, the end behavior is $-^1/_2$ as x goes to both positive and negative infinity.

$$\lim_{x\to\infty}\left(\frac{-5x^3+4x^2-10}{10x^3+3x^2+98}\right)=\lim_{x\to-\infty}\left(\frac{-5x^3+4x^2-10}{10x^3+3x^2+98}\right)=-\frac{1}{2}$$

Example 5

Translate the following limit expression into words. What do you notice about the limit expression?

$$\lim_{h\to 0}\left(\frac{f(x+h)-f(x)}{h}\right)=x$$

The limit of the ratio of the difference between f of quantity x plus h and f of x and h as h approaches 0 is x.

You should notice that $h \to 0$ does not mean h=0 because if it did then you could not have a 0 in the denominator. You should also note that in the numerator, f(x+h) and f(x) are going to be super close together as h approaches zero. Calculus will enable you to deal with problems that seem to look like $^0/_0$ and $^\infty/_\infty$.

Review

Describe the end behavior of the following rational functions at infinity and negative infinity using limits.

1. $f(x)=\frac{2x^4+4x^2-1}{5x^4+3x+9}$
2. $g(x)=\frac{8x^3+4x^2-1}{2x^3+4x+7}$
3. $f(x)=\frac{x^2+2x^3-3}{5x^3+x+4}$
4. $f(x)=\frac{4x+4x^2-5}{2x^2+3x+3}$
5. $f(x)=\frac{3x^2+4x^3+4}{6x^3+3x^2+6}$

Translate the following statements into limit notation.

6. The limit of $y=2x^2+1$ as x approaches 3 is 19.

7. The limit of $y=e^x$ as x approaches negative infinity is 0.

8. The limit of $y=^1/_x$ as x approaches infinity is 0.

Use limit notation to represent the following mathematical statements.

9. $^1/_4+^1/_{16}+^1/_{64}+\cdots=^1/_3$

10. The series $1+^1/_2+^1/_3+^1/_4+\cdots$ diverges.

11. $1+^1/_2+^1/_4+^1/_8+\cdots=2$

12. $^9/_{10}+^9/_{100}+^9/_{1000}+\cdots=1$

Translate the following mathematical statements into words.

13. $\lim_{x\to 0}\frac{5x^2-4}{x+1}=-4$
14. $\lim_{x\to 1}\frac{x^3-1}{x-1}=3$
15. If $\lim_{x\to a}f(x)=b$, is it possible that $f(a)=b$? Explain.

Review (Answers)

To see the Review answers, open this PDF file and look for section 14.1.

Vocabulary

Term	Definition
End behavior	End behavior is a description of the trend of a function as input values become very large or very small, represented as the 'ends' of a graphed function.

Term	Definition
Horizontal Asymptote	A horizontal asymptote is a horizontal line that indicates where a function flattens out as the independent variable gets very large or very small. A function may touch or pass through a horizontal asymptote.
limit notation	Limit notation is a way of expressing the fact that a function gets arbitrarily close to a value.

Additional Resource

PLIX: Play, Learn, Interact, eXplore - Concept of Limit

Video: Introduction to Limits

Practice: Concept of Limit

Real World: Suiting Up

1.2: Evaluate Limits Using Graphs

Additional Resources

PLIX: Play, Learn, Interact, eXplore - Evaluate the Limits

Practice: Evaluate Limits using Graphs

1.3: Evaluate Limits Using Tables

Additional Resources

Practice: Evaluate Limits Using Tables

1.4: Limits of Composite Functions

Limits

Did you know that you can track the "behavior" of a function? Well, you can, and I'm not talking about whether the function is making curfew or not. When we talk about the "behavior" of a function, we look at how the function "shows up" when graphed. As we solve the function, we can get closer and closer and closer - the limits helps us to understand this movement of the function on the graph.

Here are some additional resources that provide a more in-depth exploration of this topic.

- Limits and Asymptotes
- Infinite and Non-Existent Limits

Additional Resoures

Video: Find Limits of Composite Functions Graphically

Practice: Limits of Composite Functions

1.5: Evaluate Limits Using Substitution

Finding limits for the vast majority of points for a given function is as simple as substituting the number that x approaches into the function. Since this turns evaluating limits into an algebra-level substitution, most questions involving limits focus on the cases where substituting does not work. How can you decide if substitution is an appropriate analytical tool for finding a **limit**?

Using Substitution to Find Limits

Finding a limit **analytically** means finding the limit using algebraic means. In order to evaluate many limits, you can **substitute** the value that x approaches into the function and evaluate the result. This works perfectly when there are no holes or **asymptotes** at that particular x value. You can be confident this method works as long as you don't end up dividing by zero when you substitute.

If the function f(x) has no holes or asymptote at x=a then:

$$\lim_{x \to a} f(x) = f(a)$$

Occasionally there will be a **hole** at x=a. The limit in this case is the height of the function if the hole did not exist. In other words, if the function is a rational expression with factors that can be canceled, cancel the term algebraically and then substitute into the resulting expression. If no factors can be canceled, it could be that the limit does not exist at that point due to asymptotes.

Look at the following two limits:

$$\lim_{x \to 2} \frac{x^2-4}{x-2}, \quad \lim_{x \to 3} \frac{x^2-4}{x-2}$$

The limit on the left cannot be evaluated by direct substitution because if 2 is substituted in, then you end up dividing by zero.The limit on the right can be evaluated using direct substitution because the hole exists at x=2 not x=3. Thus, the limit is:

$$\lim_{x \to 3} \frac{x^2-4}{x-2} = \frac{3^2-4}{3-2} = \frac{9-4}{1} = 5$$

Examples

Example 1

Earlier, you were asked h[illegible] you should use substitution to solve a limit. In order to decide whether substitution is an appropriate first step y[illegible] ry it. You'll know it won't work if you end up trying to evaluate an expression with a denominator equal to zero. If this happens, go back and try to factor and cancel, and then try substituting again.

Example 2

Evaluate the following limit by canceling first and then using substitution.

$$\lim_{x\to 2} \frac{x^2-4}{x-2}$$

$$\begin{aligned}\lim_{x\to 2} \frac{x^2-4}{x-2} &= \lim_{x\to 2} \frac{(x-2)(x+2)}{(x-2)}\\ &= \lim_{x\to 2}(x+2)\\ &= 2+2\\ &= 4\end{aligned}$$

Example 3

Evaluate the following limit analytically:

$$\lim_{x\to 4} \frac{x^2-x-12}{x-4}$$

$$\begin{aligned}\lim_{x\to 4} \frac{x^2-x-12}{x-4} &= \lim_{x\to 4} \frac{(x-4)(x+3)}{(x-4)}\\ &= \lim_{x\to 4}(x+3)\\ &= 4+3\\ &= 7\end{aligned}$$

Example 4

Evaluate the following limit analytically.

$\lim_{x\to 3} \frac{x^2-9}{x-3}$

$\lim_{x\to 3} \frac{x^2-9}{x-3} = \lim_{x\to 3} \frac{(x-3)(x+3)}{(t-3)} = \lim_{x\to 3}(x+3) = 6$

Example 5

Evaluate the following limit analytically.

$$\lim_{y\to 4} \frac{3|y-1|}{y+4}$$

$$\lim_{y\to 4} \frac{3|y-1|}{y+4} = \frac{3|4-1|}{4+4} = \frac{3\cdot 3}{8} = \frac{9}{8}$$

Review

Evaluate the following limits analytically.

1. $\lim_{x\to 5} \frac{x^2-25}{x-5}$
2. $\lim_{x\to -1} \frac{x^2-3x-4}{x+1}$
3. $\lim_{x\to 5} \sqrt{5x}-12$
4. $\lim_{x\to 0} \frac{x^3+3x^2-x}{5x}$
5. $\lim_{x\to 1} \frac{3x|x-4|}{x+1}$
6. $\lim_{x\to 2} \frac{x^2+5x-14}{x-2}$
7. $\lim_{x\to 1} \frac{x^2-8x+7}{x-1}$
8. $\lim_{x\to 0} \frac{5x-1}{2x^2+3}$
9. $\lim_{x\to 1} 4x^2-2x+5$

10. $\lim_{x \to 0} \frac{x^2 + 5x}{x}$

11. $\lim_{x \to -3} \frac{x^2 - 9}{x + 3}$

12. $\lim_{x \to 0} \frac{5x + 1}{x}$

13. $\lim_{x \to 1} \frac{5x + 1}{x}$

14. $\lim_{x \to 5} \frac{x^2 - 25}{x^3 - 125}$

15. $\lim_{x \to -1} \frac{x - 2}{x + 1}$

Review (Answers)

To see the Review answers, open this PDF file and look for section 14.4.

Vocabulary

Term	Definition
analytically	One way to evaluate a limit analytically is by substitution. Substitution is a method of determining limits where the value that x is approaching is substituted into the function and the result is evaluated.
Asymptotes	An asymptote is a line on the graph of a function representing a value toward which the function may approach, but does not reach (with certain exceptions).
Hole	A hole exists on the graph of a rational function at any input value that causes both the numerator and denominator of the function to be equal to zero.
limit	A limit is the value that the output of a function approaches as the input of the function approaches a given value.

Additional Resources

Practice: Evaluate Limits Using Substitution

Real World: The Limits of Tolerance
